FIRM STRUCTURE AND OCCUPATIONAL SORTING

Fernando Lopes
UW-Madison

Gabriel Toledo
Purdue University

Midwest Macro - Marquette University

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MOTIVATION

- Allocation of workers across and inside firms is key for productivity
- Workers sort into occupations/hierarchies as well as firms
- However, occupational/hierarchical structure is endogenous to labor market conditions

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- However, occupational/hierarchical structure is endogenous to labor market conditions

How does <u>hierarchical structure</u> affect the sorting patterns we expect to see across occupations and firms?

THIS PAPER

- **Theory** — Model of endogenous firm hierarchy and occupational sorting
  - Tractable setting: endogenous production function, heterogeneous workers inside the firm, and nontrivial sorting patterns
  - Firms choose their size, but also the quality of workers at each *layer* of the hierarchy
  - Comparing hierarchies: NAM across *layers* but PAM across *ranks*
  - Extensions: peer effects, aggregation, multi-worker layers, search frictions

THIS PAPER

- **Empirical Evidence** — German LIAB data, matched employer-employee
  - Job switchers when moving to better firms (higher leave-one out avg wage)
    - ★ Weakly higher number of subordinates
    - ★ Strongly *smaller* relative rank in the hierarchy
- **Quantification** — Closed-form mapping from data moments to parameters
  - Calibrated model matches key features of the data: sorting patterns, firm wage and size distributions
  - Counterfactual: Skill-biased technical change increase the relative importance of higher occupations; relevant for AI and firm restructuring

LITERATURE AND CONTRIBUTION

- **Sorting with multi-worker firms** (Kremer (1993), Kremer and Maskin (1996), Eeckhout and Kircher (2018), Boerma, Tsyvinski, and Zimin (2021))
  - Endogenous firm size, heterogeneous workers
- **Knowledge hierarchies** (Garicano (2000), Garicano and Rossi-Hansberg (2006))
  - Richer sorting patterns, with NAM among occupations
- **Endogenous firm structure** (Deming (2017), Adenbaum (2022), Freund (2022))

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MODEL

MODEL SETUP

- Firms of type $z \sim G(z)$, workers of type $q \sim H(q)$
- Production organized in layers indexed by $n \in [0, m]$
- Firms choose:
  - Number of layers m (hierarchy depth)
  - Quality of worker q_{nm} at each layer n
- Workers choose firm z and position n to maximize wages

BASELINE MODEL

- Importance function $a(n, m)$ ▸ Heuristic
  - higher layers are more consequential ($\partial a / \partial n > 0$),
  - but any layer's importance dilutes as the hierarchy grows ($\partial a / \partial m < 0$)
- Production function:

$$F\left(z, \{q_{nm}\}_{n=0}^{m}\right) = z \int_0^m a(n, m)\, q_{nm}^{\sigma}\, dn, \qquad \sigma < 1$$

FIRM'S PROBLEM

-

- Firm's two-stage problem:

  i. Given m, choose $\{q_{nm}\}$ to maximize operating profit

$$\pi(z,m) = \max_{\{q_{nm}\}_{n=0}^{m}} z \int_0^m a(n,m) q_{nm}^{\sigma} dn - \int_0^m w(q_{nm}) dn \tag{1}$$

  ii. On entry, choose m to maximize lifetime profit net of cost

$$\max_{m} \pi(z,m) - c(m) \tag{2}$$

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LABOR MARKET EQUILIBRIUM

- Main equilibrium objects: $w(q_{nm})$ and $\mu(z, n, m)$ ‣ Def
- FOC:

$$w'(q_{nm}) = \sigma z a(n, m) q_{nm}^{\sigma - 1}$$

- Guess (and later verify) $q_{nm} = \mu(n, m) z$

$$w(q_{nm}) = \frac{\sigma}{1 + \sigma} \frac{a(n, m) q_{nm}^{1+\sigma}}{\mu(n, m)}$$

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EQUILIBRIUM

- Functional form for quantification (results hold more generally)
  - Importance function $a(n,m) = a_0 \left(\frac{n}{m}\right)^{\alpha}$
  - Hierarchy cost: $c(m) = \kappa m^2/2$
- **Structure:**

$$m(z) = \beta\, z^{1+\sigma}$$

More productive firms build taller hierarchies — convex in z

EQUILIBRIUM

- **Assignment:** ▸ Solution

$$\mu(z,n) = \mu_0 \frac{n^{\alpha}}{\beta^{\alpha}\, z^{\alpha(1+\sigma)-1}}$$

  - Fix n: Higher μ at lower $z \rightsquigarrow$ NAM across firms at given layer
  - Fix z: Higher μ at higher $n \rightsquigarrow$ PAM within the firm

- Average worker quality $\bar{q}(z) \propto z^{2+\sigma}$ is increasing (PAM on average)

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COMPARING HIERARCHIES

- When would the same worker be hired in two different firms?

$$\mu(z, n) = \mu(z', n') \iff \frac{n'}{n} = \left(\frac{z'}{z}\right)^{\frac{\alpha(1+\sigma)-1}{\alpha}}$$

  - Worker gains subordinates moving to better firm: NAM on layers

- Define rank $\varphi = n/m$ (relative position in hierarchy):

$$\mu(z, n, m) = \mu(z', n', m') \iff \frac{\varphi}{\varphi'} = \left(\frac{z'}{z}\right)^{\frac{1}{\alpha}}$$

  - Worker falls in relative rank moving to better firm: PAM on ranks

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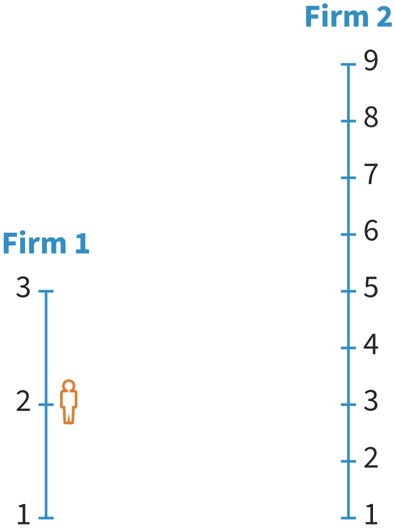
WHY STRUCTURE MATTERS



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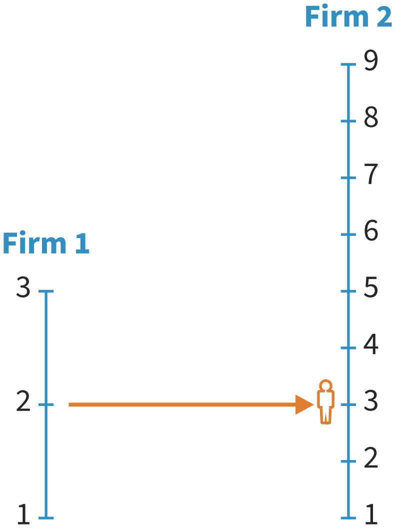

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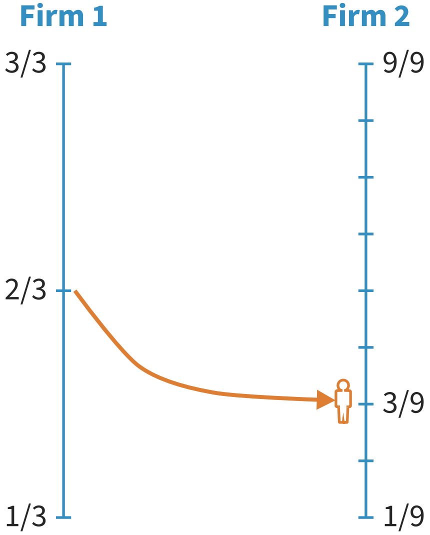

EXTENSIONS

- **Peer effects** — production complementarities within the firm
  - Stronger peer effects exacerbate PAM-ness in ranks
- **Aggregation** — model's micro sorting is consistent with macro aggregates
  - Under Pareto z: $Y = AL$ (representative-firm formulation)
- **Multi-worker layers** — Allow for more than one worker per layer; ‣ Extension
- **Directed search** — Submarkets by firm quality z and layer n ‣ Extension

EMPIRICAL EVIDENCE

LIAB DATA

- German matched employer-employee data (LIAB), yearly panel 2010–2017 (Dauth and Eppelsheimer (2020))
- West Germany, private sector, full-time workers aged 20–65
- Full workforce composition observed for each establishment in each year
- **Layer:** number of co-workers with strictly lower wages (count of subordinates)
- **Rank:** layer / firm size (relative position in the hierarchy, $\varphi \in [0, 1]$)
- **Firm quality** z_{jt}**:** leave-one-out mean log wage in the establishment

JOB SWITCHERS: REGRESSION DESIGN

- Focus on workers switching firms from one year to the next
- For worker i moving from firm j to j' at time t:

$$\Delta \log \phi_{ijt} = \gamma_0 + \gamma_1 \Delta \log z_{jt} + \text{Controls} + \varepsilon_{ijt}$$

where $\phi_{ijt} \in \{\text{Layer}_{ijt}, \text{Rank}_{ijt}\}$

- Controls: firm size, age, tenure, occupation, industry, location, year fixed effects

	Layer	Layer Bin	Rank
$\Delta \log z$	0.143	0.065	-0.915
(p-value)	(0.471)	(0.473)	(0.000)
N	7,177	7,177	7,177

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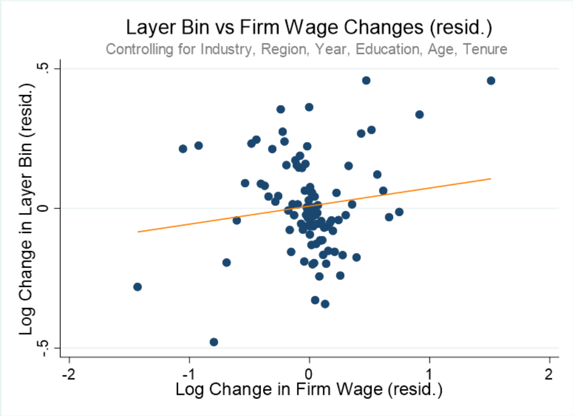
Residualized binscatter: layer



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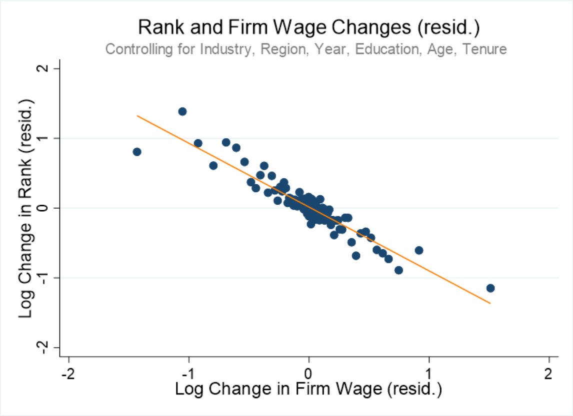


Layer: flat/positive slope — NAM on layers

Rank: negative slope — PAM on ranks

QUANTIFICATION

IDENTIFICATION

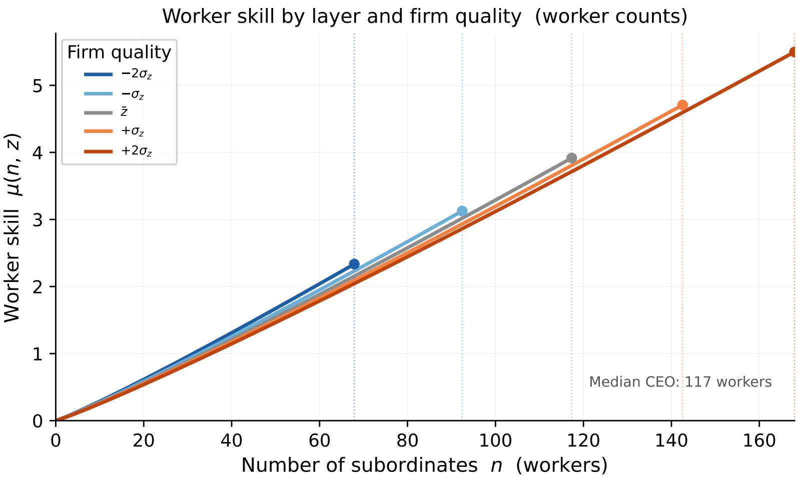
- The model delivers closed-form predictions for each empirical moment

Moment	Theory	Empirical
Rank regression coefficient	$-1/\alpha$	-0.915
Layer regression coefficient	$(1+\sigma) - 1/\alpha$	0.143
Intercept in $\log m$ on $\log z$ regression	β	0.119
$\mathbb{E}(z^{2+\sigma})$	Market clearing	17.84

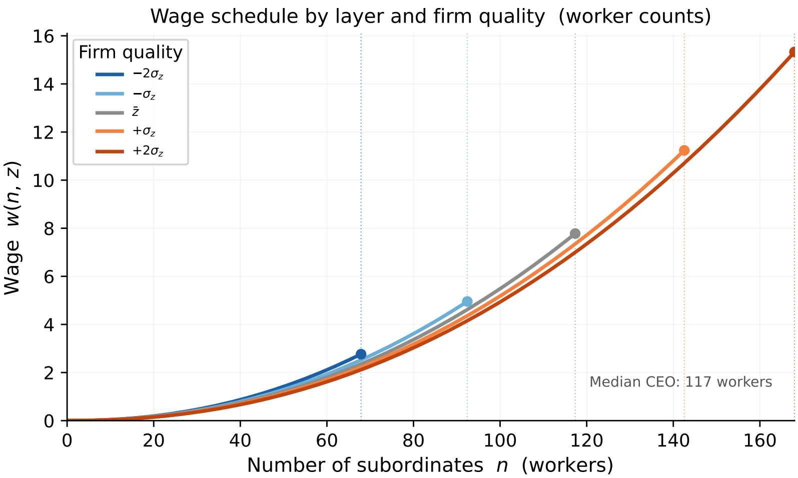
BASELINE CALIBRATION

Parameter	Symbol	Value
Skill return	σ	0.058
Importance elast.	α	1.093
Structure cost	κ	3.686
Equilibrium Assignment scale	μ_0	0.987

ASSIGNMENT ACROSS FIRMS



WAGE SCHEDULE ACROSS FIRMS



COUNTERFACTUAL: SKILL-BIASED TECHNOLOGICAL CHANGE

- Counterfactual exercise: increase α, recompute equilibrium assignment and wages, trace how sorting patterns and wage inequality respond
- Increase in α captures in the model the rise in the relative importance of higher occupations due to skill-biased technical change
- Relevant for AI and how it may reshape firm hierarchies and wage structures:

> *The overall equilibrium is a smaller industry by headcount, more concentrated at the senior level, with a hollowed-out middle.*
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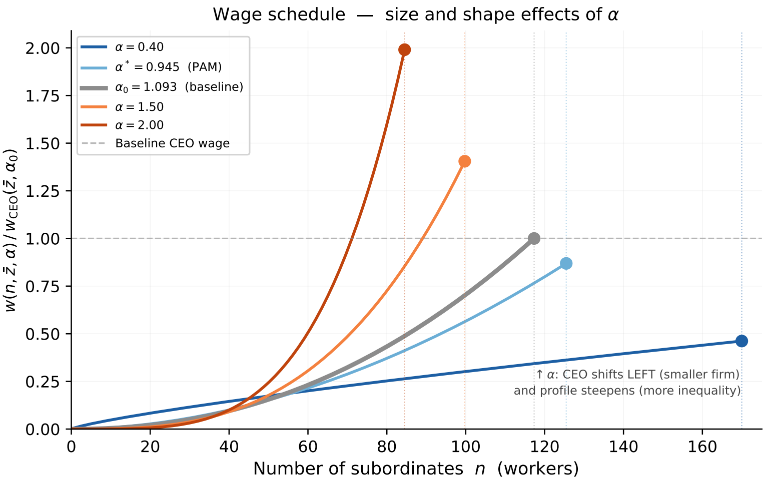
COUNTERFACTUAL: WAGE FUNCTION ACROSS α



WAGE INEQUALITY: SONG ET AL. (2019) DECOMPOSITION

	Var(log wages)				
α	Within	Between	Total	CEO / p50	Δ Within %
0.50	0.639	0.184	0.823	2.0×	−79.1
0.80	1.637	0.184	1.820	3.0×	−46.4
0.95 *	2.285	0.184	2.468	3.7×	−25.2
1.09 †	**3.054**	**0.184**	**3.238**	**4.6**×	— — —
1.50	5.754	0.184	5.938	8.0×	+88.4
2.00	10.229	0.184	10.413	16.0×	+234.9

- Within-firm wage dispersion scales as α^2: SBTC raises within-firm inequality by +88% from baseline $\alpha_0 = 1.09$ to $\alpha = 1.5$; between-firm inequality is unchanged

CONCLUSION

- Workers sort simultaneously across firms *and* within the firm hierarchy
- **Theory:** endogenous hierarchy
  - NAM on layers: at a given layer, better firms hire lower-skill workers
  - PAM on ranks: workers moving to better firms occupy lower relative positions
- **Empirics:** LIAB job switchers confirm both patterns; strong PAM on relative wage ranks
- **Quantification:** Closed-form identification, matching regression based moments and moments from firm avg wage distribution;
- **Counterfactual:** Skill-biased technical change (rise in α) leads to more concentrated hierarchies and higher within-firm wage inequality

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Thank You!

fllopes@wisc.edu

gstoledo@purdue.edu

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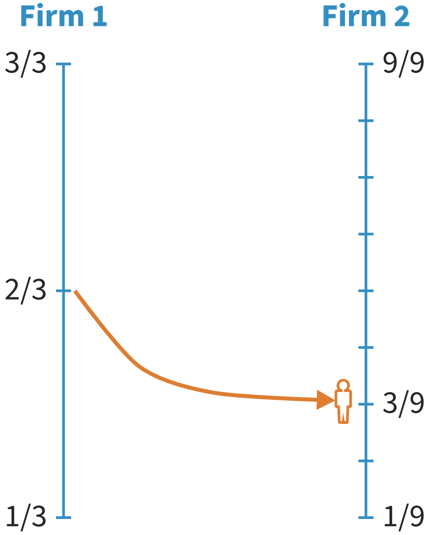
Appendix

COMPARING HIERARCHIES



Back

DEFINITION OF EQUILIBRIUM

Definition

A competitive equilibrium is a set of functions $\mu : \mathcal{Z} \times \mathbb{R} \times \mathbb{R} \to \mathcal{Q}$, $w : \mathbb{Q} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, and $m : \mathcal{Z} \to \mathbb{R}$ such that

i. μ solves (2)

ii. m satisfies (3)

iii. w satisfies worker optimality condition:

$$w(q, n, m) \geq w(q, n', m'), \ \forall (n', m') \text{ s.t. } n' \leq m' = m(z), \text{ for some } z \in \mathcal{Z}$$

iv. the allocation is feasible:

$$\int_{z}^{\bar{z}} \int_{0}^{m(z)} dndG(z') = \int_{0}^{m(z)} \int_{\mu(z,n,m)}^{\bar{q}} dH(q')dn, \ \forall z \in \mathcal{Z}$$

Back

PEER EFFECTS

- Suppose production features within-firm complementarities:

$$f(z, q_{nm}) = z \left(\int_0^m q_{nm} \, dn \right)^{\gamma} q_{nm}^{\sigma}$$

- Stronger peer effects ($\uparrow \gamma$) exacerbate PAM-ness in ranks:
  - Workers in better firms now benefit more from their high-quality colleagues
  - The rank drop when moving to a better firm is larger in magnitude

DIRECTED SEARCH

- Firms post wages; workers directedly search for jobs
- Let $\lambda(z, n, m)$ be market tightness and $p(\lambda)$ the job-finding probability
- Firms solve:

$$\max \int_0^m p(\lambda(z, n, m)) \left(a(n, m) f(z, q_{nm}) - w(q_{nm})\right) dn$$

- Workers solve:

$$\max_{(z,n,m)} \frac{p(\lambda(z, n, m))}{\lambda(z, n, m)} w(q_{nm})$$

- Supermodularity condition becomes potentially stronger than in the competitive case

MICROFOUNDATION FOR $a(n, m)$

- Law firm: two workers (lawyer, paralegal), each with 1/3 units of time
- Cases have random profile $x \sim \mathcal{U}[0, 1]$
- Optimal allocation: accept cases $x \geq 1/3$; assign $x \in [1/3, 2/3]$ to paralegal, $x \geq 2/3$ to lawyer
- Avg productivity: paralegal = 1/2, lawyer = 5/6
- When a senior lawyer joins: accept all cases; participation shares shift
  - Paralegal: $1/2 \to 1/6 \rightsquigarrow \partial a/\partial m < 0$
  - Lawyer: $5/6 \to 1/2 \rightsquigarrow$ lower layers diluted when hierarchy grows

▸ Back

SOLUTION ALGORITHM

- Define $\varphi \equiv n/m$; rewrite the worker assignment problem in rank space:

$$\max_{\{q_\varphi\}_{\varphi=0}^{1}} \int_0^1 a(\varphi)\ f(z, q_\varphi)\, d\varphi - \int_0^1 w(q_\varphi)\, d\varphi$$

- FOC: $w'(\mu(z,\varphi)) = a(\varphi)\, f_q(z, \mu(z,\varphi))$
- Market clearing implies:

$$\frac{\partial \mu(z,\varphi)}{\partial z} = \varphi\, \frac{g(z)}{h(\mu(z,\varphi))}$$

- Labor market equilibrium: family of ODEs $\{w_\varphi, \mu_\varphi\}_{\varphi\in[0,1]}$

$$\frac{dw_\varphi}{dz} = a(\varphi)\ f_q(z,\mu_\varphi)\,\frac{d\mu_\varphi}{dz}, \qquad \frac{d\mu_\varphi}{dz} = \varphi\,\frac{g(z)}{h(\mu_\varphi)}$$

PROPOSITION 3.3 (FULL STATEMENT)

Proposition

Suppose $a(n, m) = \tilde{a}(n/m)$, where $\tilde{a}(\cdot)$ is a polynomial in n/m. Then the allocation exhibits PAM if

$$\left|\varepsilon_{a,m}\right| < \frac{1}{1 + \sigma}$$

For $a(n, m) = a_0(n/m)^{\alpha}$, this reduces to $\alpha < 1/(1 + \sigma)$.

- Intuition: $a(n, m)$ must be "concave enough" in m — importance of high layers cannot rise too steeply as hierarchy expands

Back